

CELLE, Germany

WMO: 103430

Lat: 52.5958N

Lon: 10.0294E

Elev: 172

StdP: 14.60

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.9	18.4	9.5	9.0	15.9	13.7	11.1	20.6	25.3	47.0	22.8	46.6	7.1	90	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	87.4	67.0	83.6	65.4	80.0	64.1	68.7	82.4	67.1	79.8	65.5	76.8	8.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.4	91.1	71.7	62.7	85.7	71.0	61.0	80.7	69.9	32.8	82.8	31.6	79.9	30.4	77.0	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	17.7	15.5	DB	9.2	93.7	5.9	3.6	4.9	96.4	1.5	98.5	-1.9	100.5	-6.1	103.2	(4)
(5)			WB	9.0	72.1	6.5	1.9	4.3	73.5	0.5	74.7	-3.1	75.8	-7.8	77.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.0	34.6	36.3	41.3	49.2	57.0	62.0	66.3	65.4	58.3	50.3	42.0	36.9
	DBStd	13.18	8.53	8.04	6.66	7.21	6.91	6.30	6.15	5.81	5.65	6.50	6.76	8.31
	HDD50	1997	479	386	277	102	15	0	0	0	3	76	250	409
	HDD65	5705	943	804	735	474	263	133	57	66	215	455	689	871
	CDD50	2018	1	2	7	79	232	360	507	478	251	87	11	3
	CDD65	248	0	0	0	0	15	42	99	79	13	0	0	0
	CDH74	2495	0	0	0	15	185	436	1000	732	123	4	0	0
	CDH80	771	0	0	0	0	34	117	356	241	23	0	0	0
Wind	WSAvg	6.8	8.0	7.9	7.8	6.8	6.6	6.3	6.2	5.8	5.9	6.3	6.7	7.6
Precipitation	PrecAvg	28.4	2.7	2.0	2.0	1.7	2.2	2.3	2.9	2.8	2.3	2.4	2.4	2.6
	PrecMax	39.1	5.3	4.2	4.9	3.5	7.0	4.4	7.4	6.0	6.5	6.8	4.9	6.1
	PrecMin	20.7	0.4	0.6	0.3	0.3	0.7	0.9	1.0	0.7	0.4	0.7	0.0	0.8
	PrecStd	4.3	1.4	1.0	1.2	0.7	1.4	1.0	1.5	1.3	1.3	1.2	1.2	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	57.6	65.6	76.3	84.1	88.7	92.8	91.5	83.6	71.8	60.2	56.4
		MCWB	50.3	48.2	52.3	58.3	63.7	68.6	67.9	67.9	64.6	61.1	55.4	51.0
	2%	DB	51.7	53.0	60.1	71.2	79.0	83.2	88.1	86.2	77.4	66.3	57.0	53.6
		MCWB	47.4	47.0	49.7	55.8	62.0	65.7	67.0	66.0	63.0	58.8	52.6	49.9
	5%	DB	48.5	50.2	55.5	67.0	75.0	78.9	84.0	81.9	72.5	62.8	54.2	50.9
		MCWB	45.2	45.3	47.7	53.8	59.2	63.6	65.8	65.2	61.0	56.6	50.3	47.5
	10%	DB	46.1	47.8	52.0	62.8	70.9	75.0	79.4	77.3	68.5	60.1	51.9	48.1
		MCWB	43.1	43.9	46.1	52.1	57.5	61.6	64.3	63.4	59.4	55.2	48.2	45.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.0	50.4	53.6	59.9	67.0	70.4	70.7	70.2	66.7	62.1	56.0	52.4
		MCDB	53.8	54.4	62.7	72.7	80.2	83.2	84.9	84.6	77.6	69.6	58.9	55.3
	2%	WB	48.1	48.1	51.0	57.2	63.8	67.2	68.7	68.2	64.2	59.6	53.2	50.4
		MCDB	51.5	51.7	57.5	67.9	75.2	80.2	83.2	81.2	74.5	65.3	56.2	52.9
	5%	WB	45.4	45.8	48.8	55.1	61.0	65.0	67.2	66.6	62.3	57.5	50.9	47.9
		MCDB	48.1	49.7	54.4	64.5	72.0	76.0	80.7	78.0	70.5	61.9	53.9	50.7
	10%	WB	43.0	43.5	46.5	53.2	58.7	63.0	65.5	64.9	60.5	55.6	48.8	45.1
		MCDB	45.4	47.2	51.4	61.8	68.1	72.4	77.1	74.9	67.4	59.5	51.4	47.3
Mean Daily Temperature Range	5% DB	MDBR	8.7	11.1	14.3	18.2	19.2	18.5	18.6	18.8	16.9	13.9	9.6	8.5
		MCDBR	10.8	14.8	21.4	24.9	25.3	24.9	25.8	25.7	23.3	17.4	11.8	9.6
		MCWBR	8.7	10.1	13.0	12.8	11.8	11.1	9.8	10.4	11.4	10.8	8.6	8.1
	5% WB	MCDBR	10.1	12.4	18.5	21.4	23.0	22.6	22.6	21.8	19.8	16.3	11.1	9.6
MCWBR		8.7	9.3	11.9	11.7	11.5	11.0	9.5	9.9	10.5	10.6	8.7	8.6	
Clear-Sky Solar Irradiance	taub	0.318	0.338	0.372	0.411	0.412	0.410	0.426	0.419	0.386	0.357	0.336	0.313	
	taud	2.370	2.357	2.282	2.218	2.256	2.281	2.246	2.281	2.362	2.429	2.405	2.363	
	Ebn at Noon	213	243	256	260	266	268	260	255	250	234	203	192	
	Edh at Noon	19	25	33	40	41	40	41	37	31	23	18	16	
All-Sky Solar Radiation	RadAvg	202	423	763	1258	1548	1659	1571	1322	968	549	246	156	
	RadStd	20	59	79	150	173	129	213	103	114	56	27	13	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (47 neighbors)		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	-232	N/A	N/A	

Nomenclature: See separate page